Amritpal Singh

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Skills

- **Languages and Tools:** C, C++, Python, HTML, CSS, Javascript, x86_64 Assembly, Bash, GDB, CMake, GNU Make, Git, GNU Emacs, VSCode, vim, neovim
- **Other Relevant information:** Custom Linux Kernel, Self Hosted services, IoT, Homelab, Publishing and maintaining a package on AUR.

Education

I.K.G. Punjab Technical University, B.Tech in Computer Science and Engineering CGPA: 7.4 – April 2027,

Open-Source Experience

GNU Emacs (Code Editor) Commit: 46b6d1

- Bug was caused by inappropriate gzip stream end checking. An assumption was made which is always true in GNU implementation but wasn't in pigz. Therefore, I contributed a simple check that resolved the issue.

slash_tmp (Temporary File Host Server) Pull Request List

- Pull requests made contributing Key Features such as fixing inappropriate payload size in header, also enable SSL (HTTPS) support in the project. Enable chunking of files in the server so it does not load the complete file into memory causing crashing. Also implemented linux-only signal handling so it behaves like a proper server program.

TLP (Laptop Power Management Utility) Commit: 9f3ff4

- Fixed a failure in startup process of TLP. It used a timeout flag in its startup program which is non-POSIX compliant. It fails to start in non-GNU environments. Therefore I replaced the startup checks with portable options so it works as intended.

Projects

FORTH86 x86_64 assembly, CMake

- It is a FORTH implementation, written as a post-reading exercise, a complete interpreter implementation with turing complete words that fully allow the user to make use of a FORTH-like language with a focus on performance, written in raw assembly, including a self made standard library and re-implementation of common libc routines in a compact manner.

sinkhole (wip) C++, C, x86 assembly, Make

- A small kernel that outputs to a BIOS VGA Buffer and Serial output. It is capable of detecting Physical RAM and the largest usable chunks in it. It also places an IDT and a GDT in memory It also implements stack smashing protections, and rudimentary C++ support, providing for a rich base that can be expanded upon.

perfmode C, Make

- A Linux CLI Tool that allows for controlling performance modes and fan profiles provided by ASUS Notebook Kernel modules, and ACPI Platform Profile (modern), while maintaining compatibility with third party drivers such as faustus. This project was one of the initial projects that allowed for this functionality for the user in an easy manner on ASUS Laptops, as base software support with rich UI was not available at the time of authorship (circa 2020-21).

silly_packer C++, Raylib, CMake, stb

- An image packing program extremely useful in gamedev, that generates a texture atlas along with a C++ header file to use with, providing basic functionality to locate all the assets in the texture, as well as allowing for binary blob embedding. It provides optional raylib wrapper functions which allow higher level atlas loading and texture access.

c8fontgen C, Raygui, Raylib, CMake

- A small pixel editor that generates hex-code output that functions as an input for CHIP-8 fonts. The user gets to design their own font for the interpreter/emulator, with a nice UI supporting both dark and light themes, along with clipboard support.

Homelab Docker, Podman, Arch Linux

- A homelab implemented using a used Thinkpad T480 along with a 12TB capacity drive bay (filled upto 3TB), using Arch Linux as base operating system for the lab. Proxied via FRPS Tunnels behind Oracle Cloud servers. Serves media libraries using jellyfin and jellyseer, with *arr services at backend. Strong and Robust hosting of a Matrix Homeserver, allowing for secure communication at local control. Serves as DNS server using BlockyDNS.